

Contact

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nomanrazasial (LinkedIn)

Top Skills

Energy Infrastructure Modelling
Pipeline and HRS Modelling
Carbon Intensity Assessment

Languages

English (Full Professional)
Urdu (Native or Bilingual)

Certifications

Initiating and Planning Projects
IELTS Academic
Microsoft Office Specialist Word
2016

Noman Raza Sial

PhD Researcher in Sustainable Aviation Fuels | Chemical & Process
Systems Engineer | TEA, LCA & Environmental Economics
Hasselt, Flemish Region, Belgium

Summary

I am a PhD researcher with a background in Chemical and Process Systems Engineering, focused on enabling the energy transition through advanced fuel pathways, energy systems modelling, and sustainability assessment. Currently, I am pursuing a PhD at UHasselt, working on the FWO-SBO e-SAF project, which develops next-generation pathways to convert captured CO₂, water, and renewable electricity into Electro-Sustainable Aviation Fuels (e-SAF). My research focuses on identifying and quantifying the key drivers of lifecycle greenhouse gas emissions, production costs, and policy implications of novel e-SAF production pathways. Previously, my work has focused on the intersection of chemical and process engineering, techno-economic analysis (TEA), and life cycle assessment (LCA) for emerging low-carbon fuels. I have experience across hydrogen and LNG systems, including process design, system optimization, infrastructure analysis, and integrated energy systems modelling. I have also worked on the techno-economic modelling of green hydrogen production, storage, transport, and infrastructure deployment, including large-scale energy transition case studies. Core interests: Power-to-X, sustainable fuels, hydrogen systems, decarbonization of hard-to-abate sectors, integrated energy systems. Technical tools: Python, Aspen HYSYS / Aspen Plus, openLCA, and advanced data-driven modelling. Strengths: Systems thinking, techno-economic modelling, cross-disciplinary research, industry-academia collaboration. I am always interested in connecting with researchers and industry professionals working on low-carbon fuels, hydrogen, Power-to-X, and future energy systems.

Experience

Hasselt University
Doctoral Researcher
February 2026 - Present (8 months)

Hasselt, Flemish Region, Belgium

Sultan Qaboos University

1 year 1 month

Senior Researcher

July 2025 - January 2026 (7 months)

Muscat

- Developed a Python-based techno-economic model for integrated green hydrogen production, transmission, storage, and refueling systems.
- Evaluated 21 hydrogen supply pathways using hourly renewable energy simulation and infrastructure optimization.
- Conducted techno-economic and sensitivity analyses to identify key cost drivers and optimal deployment strategies.
- Provided decision support for hydrogen infrastructure planning and policy to enable freight transport decarbonization in Oman.

Senior Researcher

January 2025 - June 2025 (6 months)

Muscat

- Developed Python-based techno-economic models for integrated green hydrogen supply chains.
- Evaluated domestic infrastructure and international export pathways for renewable hydrogen.
- Assessed hydrogen production, storage, transport, and conversion into LH₂, ammonia, and methanol.
- Performed cost, lifecycle, and sensitivity analyses to identify key techno-economic drivers.
- Generated decision support for hydrogen infrastructure, trade, and energy policy.

Al-Hazm Engineering Solutions

Process Engineering Consultant

January 2024 - December 2024 (1 year)

Muscat, Masqat, Oman

- Spearheaded the development of the winning project proposal, aligning with CIRCE Biotechnologie GmbH's strategic goals.
- Identified optimal vortex tube integration points in LNG processes to boost production efficiency.
- Designed process simulations integrating vortex tubes to explore energy-saving opportunities.

- Co-developed a patent application for the innovative LNG process.

Cornell University

Visiting Graduate Researcher

December 2023 - May 2024 (6 months)

Ithaca, New York, United States

- Secured funding and led a team of 4 members to evaluate Bitcoin mining's water, land, and carbon impacts.
- Performed life cycle analysis (LCA) to measure indirect environmental footprints.
- Developed mathematical models to estimate direct environmental usage from mining facilities.
- Developed and presented monthly dashboards and presentations for effective project tracking and authored a first-author publication under review in a top-tier (Q1) journal.

Sultan Qaboos University

1 year 11 months

Senior Researcher

March 2023 - November 2023 (9 months)

Muscat, Masqaţ, Oman

- Led a team of 4 members to conduct a feasibility study for estimating the green hydrogen production potential in the unexplored north-west landlocked part of Oman.
- Supervised and made significant contributions to the development of mathematical models, monthly presentations, and final project reports.
- Proposed recommendations for Oman's national energy transition policy to boost green hydrogen infrastructure development in unexplored regions of the country.
- Published a first-author paper in a leading (Q1) journal.

Research Assistant

January 2022 - February 2023 (1 year 2 months)

Muscat, Masqaţ, Oman

- Collaborated with project partners and regional stakeholders (Oman LNG, Oman's Ministry of Energy and Minerals, Hy-Fly etc.) to assess the feasibility of Sur Hydrogen Energy Transition Cluster project.
- Developed tailored mathematical models to evaluate the techno-economic viability for the utilization of hydrogen-natural gas blends in existing infrastructure and transportation of hydrogen to other industrial cities.

- Assessed risks and developed a roadmap for hydrogen cluster installation.
- Co-authored a paper under review in a top-tier (Q1) journal.

Yeungnam University

Graduate Research Student

March 2021 - February 2023 (2 years)

Gyeongsan, North Gyeongsang, South Korea

- Conducted document review, organization, and quality control of data acquisition to draw relevant conclusions and direct research activities on liquid hydrogen and LNG systems.
- Explored decarbonization strategies for long distance transportation of hydrogen.
- Simulated, optimized, and evaluated hydrogen liquefaction processes for better energy and economic performance without increasing process complexity.
- Served as a Teaching Assistant and handled day-to-day administrative tasks for my advisor and as Systems Manager at PSDC lab at Yeungnam University.
- Contributed to four (4) Q1 papers, including one first-author publication.

Lahore Grammar School

Teacher

September 2019 - February 2021 (1 year 6 months)

Lahore, Punjab, Pakistan

- Conducted practical and tutorial sessions for Mathematics and Chemistry courses, catering to groups of up to 30 senior high school students.
- Reported on the assessment of prior curriculum, collaborated in resource development with fellow presenters, and facilitated the delivery of sessions. Also involved in coursework marking and feedback.
- Assisted in the delivery of a pre-designed practical course, guiding students in running experiments and analyzing results.

Education

Hasselt University

Doctor of Philosophy, Environmental Economics · (February 2026 - January 2030)

Yeungnam University

Master of Engineering - MEng, Chemical Engineering · (March 2021 - February 2023)

University of the Punjab

Bachelor of Engineering - BE, Chemical Engineering · (November
2014 - December 2018)